Ex. No.: 7

SELECT with various clause – GROUP BY, HAVING, ORDER BY

1. GROUP BY

How many students are registered for each course?

```
mysql> SELECT
    ->     c.COURSE_NAME,
    ->     COUNT(e.STUDENT_ID) AS TOTAL_STUDENTS
    -> FROM course c
    -> JOIN enrollment e
    -> ON c.COURSE_ID = e.COURSE_ID
    -> GROUP BY c.COURSE_NAME;
+-------------+----------------+
| COURSE_NAME | TOTAL_STUDENTS |
+-------------+----------------+
| DBMS        |              2 |
| Python      |              1 |
| AI          |              1 |
+-------------+----------------+
3 rows in set (0.03 sec)
```

2. GROUP BY

How many courses did each student register for?

```
mysql> SELECT
    ->     s.STUDENT_NAME,
    ->     COUNT(e.COURSE_ID) AS TOTAL_COURSES
    -> FROM student s
    -> JOIN enrollment e
    -> ON s.STUDENT_ID = e.STUDENT_ID
    -> GROUP BY s.STUDENT_NAME;
+--------------+---------------+
| STUDENT_NAME | TOTAL_COURSES |
+--------------+---------------+
| Rahul        |             1 |
| Priya        |             1 |
| Kiran        |             1 |
| Anjali       |             1 |
+--------------+---------------+
4 rows in set (0.01 sec)
```

3. ORDER BY

Retrieve Student Name in ascending order of Student ID

```
mysql> SELECT
    ->     STUDENT_NAME
    -> FROM student
    -> ORDER BY STUDENT_ID ASC;
+--------------+
| STUDENT_NAME |
+--------------+
| Rahul        |
| Priya        |
| Kiran        |
| Anjali       |
+--------------+
4 rows in set (0.00 sec)
```

4. ORDER BY

List the faculty members in the order of older faculty first.

```
mysql> SELECT
    ->     FACULTY_NAME,
    ->     AGE,
    ->     STATUS
    -> FROM faculty
    -> ORDER BY AGE DESC;
+--------------+------+---------+
| FACULTY_NAME | AGE  | STATUS  |
+--------------+------+---------+
| PRIYA        |   62 | WORKING |
| MOHAN        |   59 | WORKING |
+--------------+------+---------+
2 rows in set (0.00 sec)
```